What Makes a Song Popular?

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1. Abstract

This report investigates the relationship between quantified, numeric audio features and track popularity on Spotify using a dataset of more than 100,000 tracks spanning 114 genres. Our primary goal is to determine which audio characteristics are most strongly associated with a song's popularity, if such association exists, and if so, how these associations vary across features.

We used four statistical methods to observe these relationships: Pearson correlation coefficients, multiple linear regression (MLR), one-way ANOVA, and two sample t-tests. No single audio characteristic was found to be a strong linear predictor of popularity. Through various observations between audio features and popularity, it is apparent that their relationship is genre relative as feature popularity correlations varies substantially across genres.

Track duration differs significantly across genres with most mainstream genres being in the 3–4 minute range, while electronic subgenres average over 5 minutes. Another interesting observation is that mean popularity declines for tracks with 'speechiness', suggesting listeners prefer to listen to songs that are more music dominant. Finally, explicit tracks are significantly more popular overall, but this advantage reverses in hip hop and even disappears in pop, once again confirming that a song's sonic features and popularity is genre contingent with no consistent criteria for a hit song.

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3. Reason for Study

As Georgia Tech students, we love to listen to music. Whether it is to pass the time, study, fill eerie and quiet spaces, or even lighten up the mood a little bit, music has a profound impact on our day to day lives. Spotify is the world's largest music streaming platform, consisting of over 100 million tracks and 761 million active users (Spotify, 2026). Further, Spotify keeps track of each song's quantitative statistics, and attributes like popularity, tempo, and length of a song help provide useful insights into the "essence" of a song. Our group looks to determine whether or not these attributes can be used to (1) understand the spread and variability of these metrics in music, (2) determine which metrics can accurately predict the popularity of a song, and (3) identify metrics that are genre dependent.

The implications of this study are numerous. On a commercial level, producers, songwriters, and studios have been trying to understand what makes a "top hit" for years. The ability to accurately identify metrics that will lead to success enables these entities to produce more popular songs and in return make more money. Further, we live in a demand-based society where people wants and preferences drive supply. Accurately understanding what makes a song popular enables the music market to tailor their supply to the proven demand. Finally, on a human level, metrics like loudness, energy, and explicitness could potentially provide insights for future research.

This study applies Pearson correlation coefficients, multiple linear regression (MLR), one-way ANOVA, and two sample t-tests to address this aforementioned curiosity and title of our report: *What makes a song popular?*

4. Dataset

The dataset used in this report was obtained from Kaggle and contributed by Marshi Pandya, who collected and preprocessed the data using Spotify's Web API and Python. The original dataset contains 114,000 rows and 20 columns. However, one observation was removed due to a missing track name making the size n=113,999. [A] Each row in the dataset represents a unique track, identified by the song's track ID, track name, artist name, album name, and one categorical label (track genre), along with 10 numeric audio metrics, boolean explicit flag, and an integer popularity score.

These include:

- Loudness - in decibels (dB)
- Tempo - in beats per minute (BPM)
- Duration - logged in milliseconds but converted to minutes
- Danceability - combination of tempo, rhythmic stability, and beat strength (score 0 to 1)

- Energy - perceived intensity and activity (score 0 to 1)
- Speechiness - presence of spoken words (score 0 to 1)
- Acousticness - probability that the song is acoustically made (score 0 to 1)
- Instrumentalness - probability of absence of vocals (score 0 to 1)
- Liveness - probability of a live audience in recording (score 0 to 1)
- Valence - how positive the music is (score 0 to 1)
- Explicitness - True or False which we assigned a 1 or 0 for analysis
- Popularity - a score from 0 (least popular) to 100 (most popular)

5. Statistical Analyses

Table 1: Descriptive Statistics for Popularity and Audio Features (n=113,999)[B]

Attributes	Mean	Std	Min	Q1	Median	Q3	Max
Popularity	33.24	22.31	0.00	17.00	35.00	50.00	100.00
Danceability	0.567	0.174	0.000	0.456	0.580	0.695	0.985
Energy	0.641	0.252	0.000	0.472	0.685	0.854	1.000
Loudness (dB)	-8.26	5.03	-49.53	-10.01	-7.00	-5.00	4.53
Speechiness	0.085	0.106	0.000	0.036	0.049	0.084	0.965
Acousticness	0.315	0.333	0.000	0.017	0.169	0.597	0.996
Instrumentalness	0.156	0.310	0.000	0.000	0.000	0.049	1.000
Liveness	0.214	0.190	0.000	0.098	0.132	0.273	1.000
Valence	0.474	0.259	0.000	0.260	0.464	0.683	0.995
Tempo (BPM)	122.15	29.98	0.000	99.22	122.02	140.07	243.37
Duration (min)	3.801	1.788	0.143	2.901	3.548	4.358	87.29

5.1 Dataset Statistics

Table 1 shows the mean, standard deviation, min-max, and quartiles for all numerical data values. Our largest focus, popularity, is approximately symmetric around a mean of 33.24 and median of 35, with a standard deviation of 22.31. The upper quartile value of 50.00 suggests that very few tracks achieve high popularity. The max represents Sam Smith's song "Unholy" which in 2022 was the number one hit for 4 weeks. This aligns with real world behavior as very high

songs become extremely popular or viral then later die down a few weeks after stardom. Several features, notably instrumentalness, acousticness, and speechiness, are heavily right-skewed. Their medians are near their min value while their means are substantially higher indicating either (1) the scarcity of these attributes in songs or (2) large outliers that heavily skew the data. It is interesting to note that instrumentalness and acousticness also have the highest coefficient of variation among all attributes. Loudness ranges from −49.53 dB to 4.53 dB, with a mean of −8.26 dB.

5.2 Audio Features and Track Popularity

To determine if each audio feature could predict track popularity, Pearson correlation coefficients were computed for the entire dataset using the formula below. Figure 1 visualizes the computed correlation coefficients and arranges them from the least to the most positive correlation. [C]

$$\hat{\rho} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \times \sum_{i=1}^{n}(y_i - \bar{y})^2}} \qquad -1 \le \hat{\rho} \le 1$$

Formula for Pearson correlation coefficient [4]

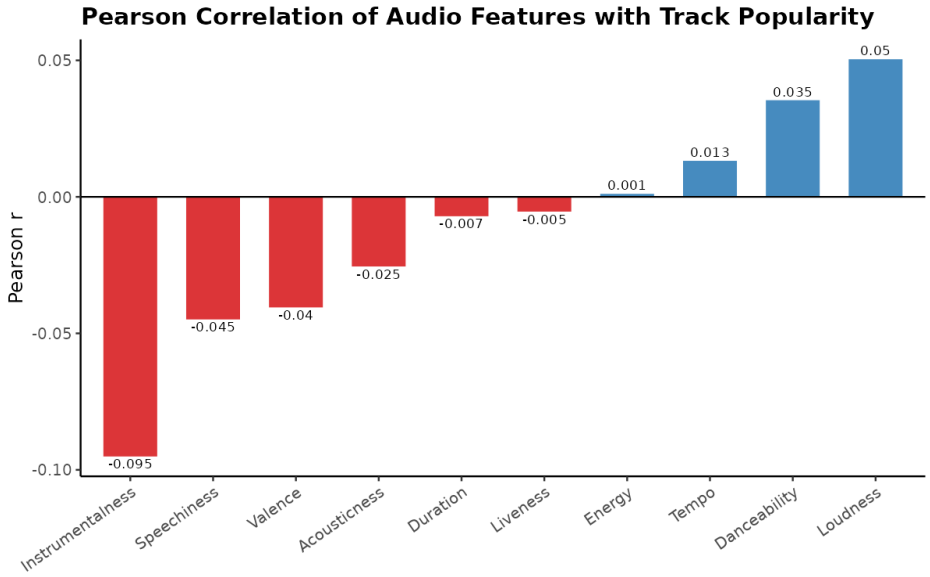


Figure 1. Pearson correlation coefficients between each audio feature and track popularity. Bars are ordered from most negative (left) to most positive (right). Red bars indicate negative correlations; blue bars indicate positive correlations.

None of the 10 features show a strong linear relationship with popularity as the greatest absolute correlation coefficient is 0.095 (instumentalness). This leads us to believe that the audio characteristics themselves do not act as reliable predictors for a song's popularity. A negative correlation score indicates that tracks with more instrumental content, typically associated with fewer vocals, tend to be slightly less popular. However, the second lowest coefficient is

spechiness. This could mean that people prefer a balance between the quantity of words in the songs that they listen to. Loudness is the strongest positive predictor, with a correlation score of r = 0.05 suggesting that louder tracks trend marginally higher in popularity. All three positive attributes are all associated with upbeat, feel-good tracks that appeal to a more mainstream audience, songs that can be used for radio and advertisements that show on TV. Surprisingly, energy is essentially uncorrelated with popularity, which fails to support the reasoning why tempo, danceability, and loudness as a group perform as positive predictors.

While all correlations are statistically significant given the very large sample size, their magnitude is too weak to create any concrete claims on the relationship between attribute and popularity. A correlation of r = 0.095 implies that instrumentalness accounts for less than 1% of the variance in popularity on its own. This finding establishes the tension of the analysis of the report. The audio features Spotify provides, taken individually, do not predict listener popularity in any applicable meaningful sense.

Multiple Linear Regression

Due to the lack of any one single attribute possessing strong predictive power, a multiple linear regression (MLR) model was fitted with popularity as the response variable to assess the collective statistical influence of all audio features together. This model allows for potential insight into the possibility that audio features may explain more variance when considered jointly than in isolation. Table 2 summarizes the model fit statistics.

An F-test was conducted to evaluate the joint significance of the attributes.

$$\begin{aligned} Popularity = {} & \beta_0 + \beta_1(Danceability) + \beta_2(Energy) + \beta_3(Loudness) \\ & + \beta_4(Speechiness) + \beta_5(Acousticness) + \beta_6(Instrumentaln\,ess) \\ & + \beta_7(Liveness) + \beta_8(Valence) + \beta_9(Tempo) + \beta_{10}(Duration) + \varepsilon \end{aligned}$$

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \cdots = \beta_{10} = 0$$

$$H_1 : \beta_j \neq 0 \text{ for at least one } j$$

$$F_0 = \frac{SSR/k}{SSE/(n-p)} = \frac{MSR}{MSE}$$

$$F_0 = 263.38 > F_{0.05,10,113988} = 1.83$$

The overall model is statistically significant ($F = 263.38$, $p < 0.001$) as it overwhelmingly rejects the null hypothesis. This confirms that the audio characteristics are statistically significant in relation to popularity. However, $R^2 = 0.023$ indicates that the joint impact only accounts for 2.3% of the variation in popularity, meaning this model cannot be used to accurately predict popularity based on these metrics alone. In practice, this model has no proper applicable value. This finding

is broadly consistent with individual correlations. MLR confirms that even when factoring in multiple characteristics, the audio metrics explain very little of what makes a track popular.

Table 2. Multiple Linear Regression: Model Statistics [D]

Metric	Value
R^2	0.0226
F-statistic	263.38
F p-value	< 0.001
n	113,999

5.3 Intercorrelations Among Audio Features

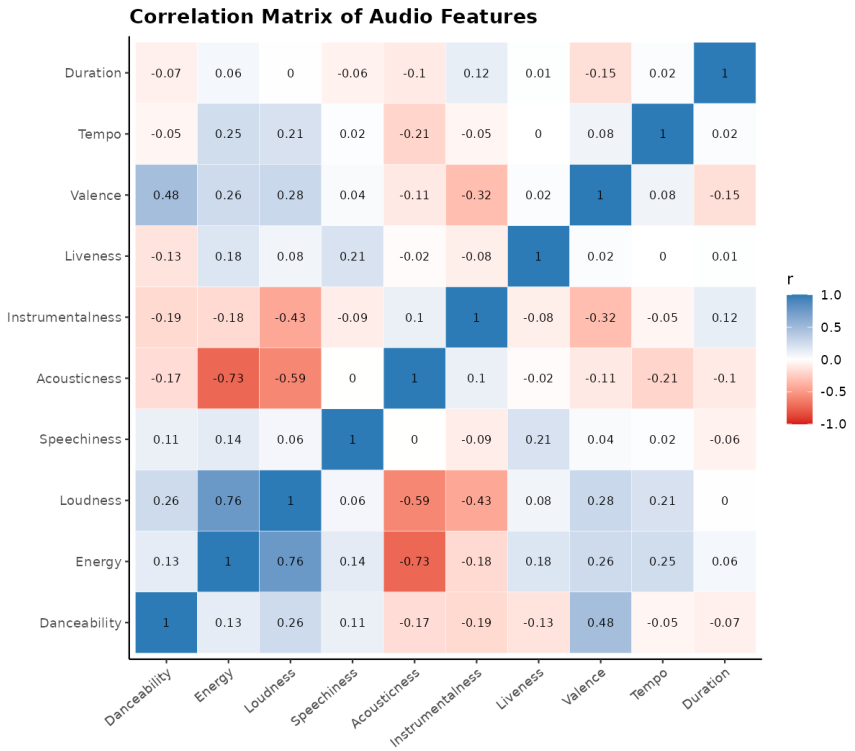


Figure 2. Pearson correlation matrix among all 10 audio features. Numerical values show the correlation coefficient r. Cells shaded blue indicate positive correlations; cells shaded red indicate negative correlations. Intensity of shading increases with the magnitude of the correlation. [E]

While the relationship between the song's metrics and popularity is the main focus of this study, it is also interesting to observe the relationship that they have with each other. Figure 2 is a correlation matrix illustrating the interconnectedness of the variables that we are working with. In the figure loudness and energy are the most positively correlated (r=0.76), meaning that there is a semi-strong linear relationship between these two metrics. Furthermore, valence and danceability

also have a moderate linear relationship (r=0.48). This means that happier sounding music likely leads to a song being perceived as better to dance to.

On the other hand, acousticness and energy show a semi-strong negative linear relationship (r=-0.73). This makes sense as acoustic songs tend to be more mellow due to the instrument choice. Finally, loudness, which is positively correlated with energy, also has a moderate negative relationship with acousticness and instrumentalness (r=-0.59 and r=-0.43). This likely means that acoustic songs are quieter and have less energy while songs with higher instrumentalness are also quieter but possess a higher amount of energy than their acoustic counterpart.

Genre Level Variation in Feature–Popularity Correlations

The weak global correlations above could not produce strong evidence to predict popularity based off audio features. We recalculated the correlation between each audio feature and popularity for every genre with more than 50 tracks, to see if there were stronger patterns hidden in genre. Figure 3 below shows that these relationships vary widely by genre. The same feature can be positively related to popularity in one genre but negative in another. For example, loudness tends to align more positively with popularity in energetic genres, while calmer genres such as sleep or study music may favor lower loudness.

This suggests that popularity is strongly genre dependent. A global analysis averages across very different musical contexts, which can hide important within-genre patterns. Therefore, audio features should not be treated as universal predictors of popularity. Their effect depends on genre norms, listener expectations, and the type of music being analyzed.

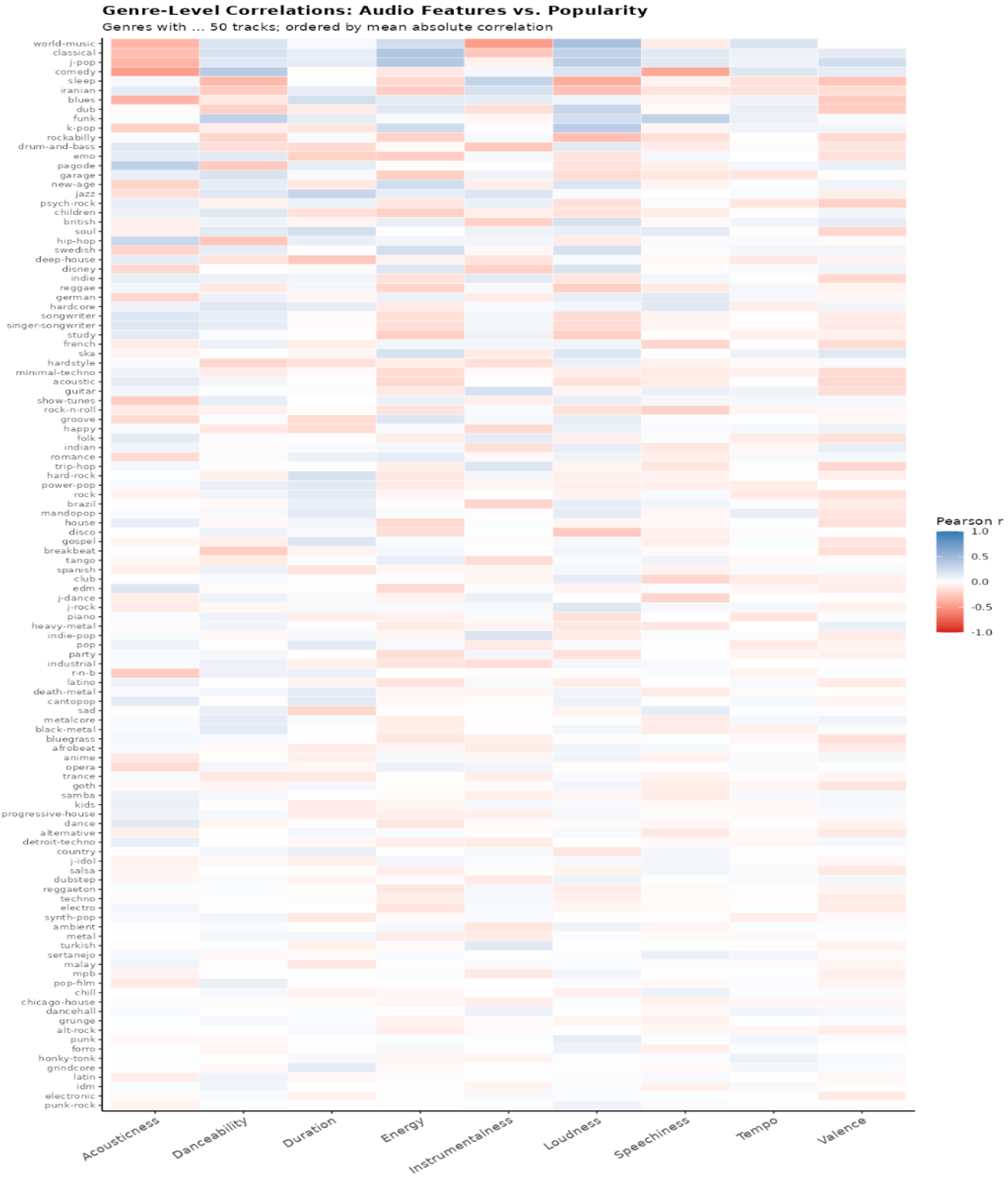


Figure 3. Genre-level Pearson correlations between each audio feature (columns) and popularity, for genres with at least 50 tracks (rows, ordered by mean absolute correlation). Blue cells indicate positive correlations; red cells indicate negative correlations. [F]

5.4 Track Duration and Popularity

One Way ANOVA: Duration Across Genres

Despite us finding little predictive strength in duration on the popularity of a song. We wanted to see whether or not duration would be impactful at the genre level. In order to do this, we conducted an ANOVA test across the 114 different music genres provided using R [G].

Table 3. One Way ANOVA: Track Duration by Genre

Source	df	SS	MS	F	p-value
Genre	113	60,002.60	530.997	198.56	< 0.001
Residuals	113,885	304,551.73	2.674		

$$R^2 = \frac{SS(genres)}{SS(total)} = \frac{60{,}002.60}{304{,}551.73 + 60{,}002.60} = 0.165$$

Based on the ANOVA test, we found statistical evidence that there is a difference in the mean track duration based on genre (p-value <0.001 for, so we reject alpha = 0.05). However, like in the past conclusions, the R-squared score is not strong (0.165). This means that there are likely minimal possible implications for this model. However, with 113,999 songs across 114 genres, this was slightly expected.

Figure 4 shows a series of 20 sequential box plots of 20 selected genres (due to either prevalence or extreme duration of time). Grindcore, a type of extreme metal, has the lowest median track duration while Minimal Techno, a genre frequently used by DJs, has the highest median track length. Furthermore, popular genres like country, hip hop, jazz, and rock tend to be in the middle of the median track durations (between 2 minutes to 4 minutes). Finally, classical has the highest interquartile range, greatest maximum, and lowest minimum meaning it has the largest gap between its song durations, and likely the greatest variance. [H]

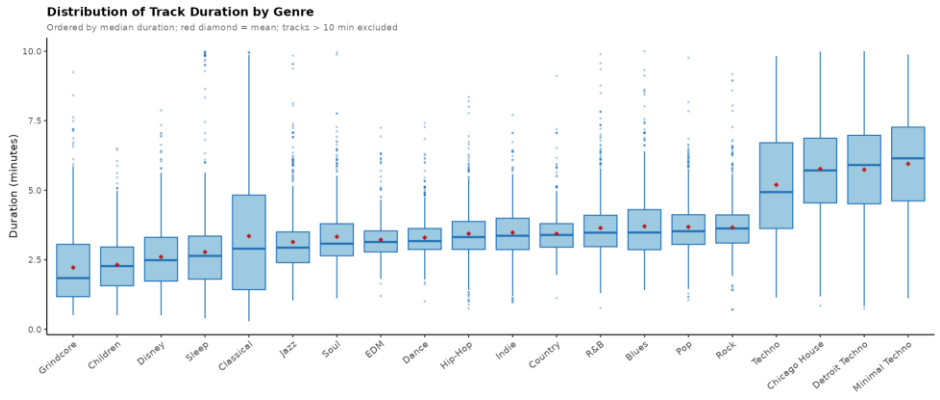


Figure 4. Boxplots of track duration for 20 selected genres (extreme durations and mainstream), ordered by median duration from shortest (left) to longest (right). Red diamonds indicate group means. Tracks exceeding 10 minutes are excluded for visualization clarity.

5.5 Speechiness and Popularity

Another important attribute of songs is its “Speechiness” or what proportion of the song contains spoken words. As shown in Table 1, the average proportion of spoken words in a song is 0.085. This means that there is typically not a large proportion of a song that is spoken. However, in order to determine if higher speechiness has a visible impact on a song's popularity, we binned tracks by intervals of 0.05 and plotted them on a bar graph (see figure 5).

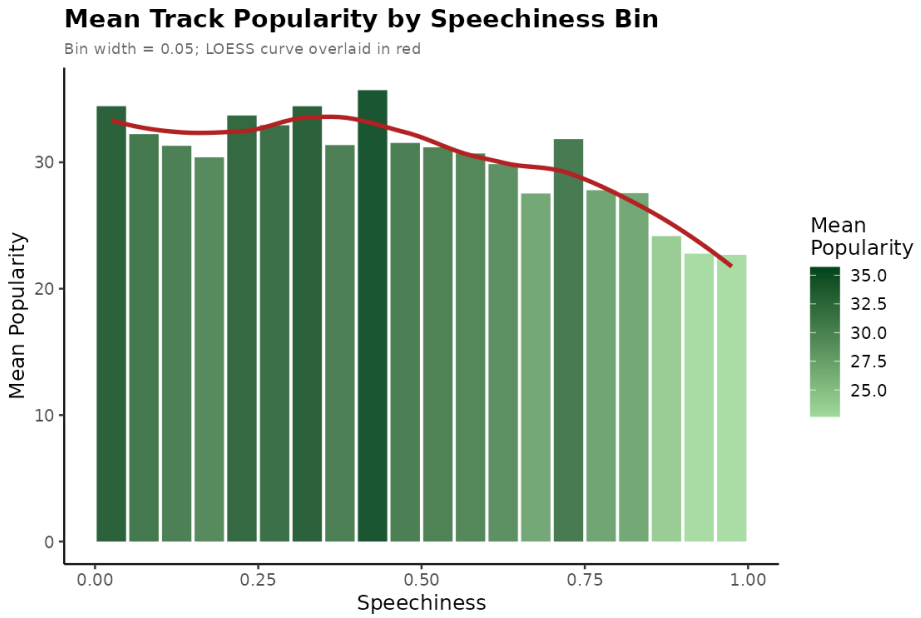


Figure 5. Mean track popularity within 0.05-wide speechiness bins. The red curve highlights the overall trend. Bars at the right end of the scale represent substantially smaller sample sizes and should be interpreted with caution.

Figure 5 illustrates with the red LOESS trend line the negative association between speechiness and a song's popularity. Songs with low or close to average speechiness hover around the mean song's popularity (33.24, Table 1). However, as the speechiness gets higher, the LOESS curve in Figure 5 dips down towards a popularity in the mid-twenties. However, since there are fewer songs with high levels of speechiness (>0.5), the data could be skewed. For instance, the bins from 0.5 to 0.7 only encompass around 100 tracks while the bins above 0.7 encompass fewer than 100 tracks. This makes their means less reliable. [I]

5.6 Explicit Content and Popularity

Two Sample t Test

Our dataset as aforementioned labels a song's explicitness as either true or false, to highlight whether the song contains profane language. Out of the 113,999 tracks in this study, 8.55% of them (9,747) were labeled as explicit. We used a two-sample t test to ascertain whether or not there was a difference in popularity between the tracks that were explicit vs those that were not. In our data we cannot assume equal variances, so we used the following hypothesis and formulas where non-explicit is μ_1 and explicit is μ_2:

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_A: \mu_1 - \mu_2 \neq 0$$

$$t_0 = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \quad [5]$$

$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1 - 1} + \frac{(S_2^2/n_2)^2}{n_2 - 1}} \quad [5]$$

Table 4. Mean Popularity by Explicit Status (Overall, n = 113,999)

Group	n	Mean Popularity	SD
Non Explicit	104,252	32.94	22.08
Explicit	9,747	36.45	24.32

Two sample t-test result with unknown variance: *t = −13.75, v (df) = 11,300.62, p < 0.001; 95% CI for (non-explicit − explicit): [−4.02, −3.01] [J]*

The summary statistics shown in Table 4 show that the mean popularity of explicit song have a higher mean than those which are not explicit (36.45>32.94). Furthermore, the explicit songs have a slightly larger standard deviation in their popularity (24.32 > 22.08). Further through the two-population t test, we found a p value less than 0.001 which proved our evidence statistically significant. Further, zero is not in the bounds of the confidence interval [-4.02, -3.01] backing up our claims. The confidence interval also shows that non-explicit songs typically fall between 3 and 4 points below explicit songs in popularity.

Genre Level Analysis

The next question is does the difference in popularity between songs with and without explicitness translate to the genres as well. To test this, we selected four genres and repeated a similar two sample t test for populations with unknown and unequal variances. The genres were Hip Hop, Pop, Country, and Comedy. Table 5 highlights the results:

Table 5. Welch Two Sample t Test: Mean Popularity by Explicit Status Across Selected Genres [K]

Genre	Mean (Non Exp.)	Mean (Explicit)	Difference	|t|	p-value	Significance
Hip Hop	44.54	23.29	−21.25	9.45	< 0.001	Yes
Pop	47.03	54.45	+7.42	1.62	0.110	No
Country	16.29	40.97	+24.68	8.21	< 0.001	Yes
Comedy	23.24	25.36	+2.12	5.41	< 0.001	Yes

Pop had the highest popularity for both explicit and non-explicit tracks (respectively 54.45 and 47.03). The lowest popularity in Table 5 was for non-explicit Country music. Finally, Hip Hop was the only genre that decreased in popularity when compared by explicitness. All changes were statistically significant except for Pop (p-score = 0.110). This is interesting as while above we found that for all tracks explicitness increases a song's popularity by around 3 to 4 points; this change in popularity is likely genre dependent. As seen with Hip Hop, the relationship between explicitness and popularity is not consistent across genres. Pop was the only genre that we could not make a definite conclusion on. Therefore, more research can be done to determine explicitness's impact on popularity in Pop.

Two results warrant closer scrutiny. Hip Hop's 21.25-point drop for explicit tracks contradicts the genre's commercial reality, and Country's 24.68-point boost is suspect given how rare explicit tracks are in mainstream country. These patterns are likely artifacts of the data: sample sizes within each genre split are imbalanced, and Spotify's single-genre tagging may pull older or international catalog tracks into "hip-hop" while inflating Country's small explicit group with a few crossover hits. A controlled comparison matching tracks on artist or release year would be needed to isolate the true effect of explicitness.

6. Conclusion

Our study looked to answer the question: *What makes a song popular?* The first part of this was understanding the 12 metrics' mean, variation, and spread. We found that average popularity scores were approximately symmetric around 33.24 or a median of 35. There was an

interconnectedness within the metrics as tracks with higher energy positively correlated with being louder and negatively correlated with acousticness. Further, valance positively correlated with danceability, and loudness negatively correlated with instrumentalness and acousticness. This likely means that energizing, upbeat, and loud music tends to not be paired with long instrument solos or nonelectrical instrument parts. This also was shown to be dependent on the type of genre the track came from as each genre's audience showed different preferences for the metrics.

Further, through the utilization of Pearson correlation coefficients we found that the greatest metric contributor to popularity, although only slightly, was loudness and danceability while the greatest detractor to popularity was instrumentalness and speechiness. This means that for artists, labels, and studios creating music, they want to focus on louder danceable beats rather than large gaps for spoken words or instrument solos. While these metrics stood out, their R score was too low to make reasonable predictions for a song's popularity (<0.1). Using MLR, we also found that when taking into account all of these variables together, they were statistically significant when determining popularity, but similarly the R squared score was 0.023, too low to reasonably predict popularity. Looking more into individual metrics, we determined using ANOVA that track duration did statistically differ based on genre, but that no predictions could be made purely by track duration. Further, most of the popular genres tended to be on average in the middle range (2-4 minutes) of the durations of tracks. Speechiness was found to be inversely correlated with popularity. However, since most songs typically have a low amount of speechiness (0.08), this had a limited impact. Finally, explicit tracks in general were shown to be about 3 to 4 points more popular than non-explicit songs. However, this was shown to be genre dependent as genres like Hip Hop saw a potentially negative influence on popularity due to explicitness.

These findings have a few implications. The first is for those in the music industry as this study provides feedback on what works and what does not to a limited extent. Using this research, an entity cannot actively predict how popular their song will be as there are likely various external factors also in play. However, this study does show that artists can lean into loudness and energy to potentially give their music a popularity boost, or they can stray away from acousticness or instrumentalness to prevent it from losing an audience. Artists should potential stay away from spoken words in their lyrics and try to keep the duration of their track between 2 to 4 minutes. However, this is also genre dependent and can lead to more research for certain genres. This retrospective assessment has the possibility of making an artist's songs more popular, increasing recognition, money, and partnerships. Further, these insights have the possibility to lead into a retrospective assessment of humans their tastes and preferences and their psychology as a whole.

Further Questions

Future research can continue or study by delving into a variety of topics. First, they can look into how different genres themselves are impacted by these metrics since we only did a surface level assessment. Further, metrics like explicitness can be delved deeper, and analyzed in how they are impacted by people's perception of the genre they are in. Another potential research topic is gauging how human have impacted these metrics: how attention span impacts song duration, how speechiness is perceived as bad, and why humans prefer louder songs.

References

[1] Pandya, M. (2022). “Spotify Tracks Dataset”. *Kaggle.* Retrieved from https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset

[2] Spotify. (2026). “About Spotify”. *Spotify Newsroom.* Retrieved April 29, 2026 from

https://newsroom.spotify.com/company-info/

[3] Spotify for Developers. (n.d.). “Audio Features Object”. *Spotify Web API Reference*

[4] Yousefi Zowj, P. (n.d) “Linear Regression (Chapter 11)”

[5] Yousefi Zowj, P. (n.d) “Hypothesis Testing 2-Pop"

Appendix

[A] Data Loading & Cleaning:

```r
df <- read.csv("dataset.csv", stringsAsFactors = FALSE)

# Drop stray index column and the one row with missing track name
df <- df[, !names(df) %in% "X"]
df <- df[df$track_name != "" & !is.na(df$track_name), ]

# Convert duration to minutes; explicit to logical
df$duration_min <- df$duration_ms / 60000
df$explicit     <- df$explicit == "True"

audio_features <- c("danceability", "energy", "loudness", "speechiness",
                    "acousticness", "instrumentalness", "liveness",
                    "valence", "tempo", "duration_min")

cat("Rows:", nrow(df), "| Genres:", length(unique(df$track_genre)), "\n")
```

[B] Table 1: Descriptive Statistics:

```r
desc_vars <- c("popularity", audio_features)

desc_tbl <- do.call(rbind, lapply(desc_vars, function(v) {
  x <- df[[v]]
  data.frame(
    Variable = v,
    N        = length(x),
    Mean     = round(mean(x, na.rm = TRUE), 3),
    SD       = round(sd(x, na.rm = TRUE), 3),
    Min      = round(min(x, na.rm = TRUE), 3),
    Q1       = round(quantile(x, .25, na.rm = TRUE), 3),
    Median   = round(median(x, na.rm = TRUE), 3),
    Q3       = round(quantile(x, .75, na.rm = TRUE), 3),
    Max      = round(max(x, na.rm = TRUE), 3)
  )
}))
```

[C] Figure 1 / Pearson Correlations with Popularity:

```
cor_vals <- sapply(audio_features, function(f) {
  cor(df[[f]], df$popularity, method = "pearson", use = "complete.obs")
})

cor_df <- data.frame(
  Feature     = names(cor_vals),
  Correlation = round(as.numeric(cor_vals), 4)
) %>% arrange(Correlation)

write.csv(cor_df, "table2_feature_popularity_correlations.csv", row.names = FALSE)
cat("Table 2 saved.\n")
```

[D] Table 2: Multiple Linear Regression:

```
mlr_formula <- as.formula(paste(
  "popularity ~", paste(audio_features, collapse = " + ")
))
mlr_model <- lm(mlr_formula, data = df)
mlr_sum   <- summary(mlr_model)

mlr_stats <- data.frame(
  Metric = c("R-squared", "Adjusted R-squared", "F-statistic", "F p-value", "n"),
  Value  = c(
    round(mlr_sum$r.squared, 4),
    round(mlr_sum$adj.r.squared, 4),
    round(mlr_sum$fstatistic[1], 2),
    format(pf(mlr_sum$fstatistic[1], mlr_sum$fstatistic[2],
              mlr_sum$fstatistic[3], lower.tail = FALSE), scientific = TRUE),
    nrow(df)
  )
)
```

[E] Figure 2: Audio Feature Correlation Matrix:

```
feat_matrix_vars <- c("danceability", "energy", "loudness", "speechiness",
                      "acousticness", "instrumentalness", "liveness",
                      "valence", "tempo", "duration_min")

cor_matrix <- cor(df[, feat_matrix_vars], use = "complete.obs")

feat_labels <- c(danceability = "Danceability", energy = "Energy",
                 loudness = "Loudness", speechiness = "Speechiness",
                 acousticness = "Acousticness", instrumentalness = "Instrumentalness",
                 liveness = "Liveness", valence = "Valence",
                 tempo = "Tempo", duration_min = "Duration")

colnames(cor_matrix) <- feat_labels[feat_matrix_vars]
rownames(cor_matrix) <- feat_labels[feat_matrix_vars]

cor_melt       <- melt(cor_matrix)
names(cor_melt) <- c("Var1", "Var2", "Correlation")
cor_melt$Var1  <- factor(cor_melt$Var1, levels = feat_labels)
cor_melt$Var2  <- factor(cor_melt$Var2, levels = feat_labels)

p_corrmat <- ggplot(cor_melt, aes(x = Var1, y = Var2, fill = Correlation)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Correlation, 2)), size = 2.8, color = "black") +
  scale_fill_gradient2(low = "#d7191c", mid = "white", high = "#2c7bb6",
                       midpoint = 0, limits = c(-1, 1), name = "r") +
  labs(title = "Correlation Matrix of Audio Features",
       x = NULL, y = NULL) +
  theme_classic(base_size = 11) +
  theme(axis.text.x = element_text(angle = 40, hjust = 1),
        plot.title  = element_text(face = "bold"))
```

[F]Figure 3: Genre-Level Correlation Heatmap:

```
genres_50 <- df %>% count(track_genre) %>% filter(n >= 50) %>% pull(track_genre)

genre_cor_df <- df %>%
  filter(track_genre %in% genres_50) %>%
  group_by(track_genre) %>%
  summarise(across(all_of(audio_features),
                   ~ cor(.x, popularity, use = "complete.obs"))) %>%
  pivot_longer(-track_genre, names_to = "feature", values_to = "r")

genre_order <- genre_cor_df %>% group_by(track_genre) %>%
  summarise(m = mean(abs(r))) %>% arrange(desc(m)) %>% pull(track_genre)

genre_cor_df$track_genre <- factor(genre_cor_df$track_genre, levels = genre_order)

ggplot(genre_cor_df, aes(feature, track_genre, fill = r)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "#d7191c", high = "#2c7bb6", midpoint = 0,
                       limits = c(-1, 1), name = "Pearson r") +
  labs(x = NULL, y = NULL) +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 40, hjust = 1),
        axis.text.y = element_text(size = 6))
```

[G] ANOVA:

```r
aov_dur       <- aov(duration_min ~ track_genre, data = df)
aov_dur_tidy  <- tidy(aov_dur) %>%
  mutate(across(where(is.numeric), ~ round(., 4)))

write.csv(aov_dur_tidy, "table4_anova_duration_by_genre.csv", row.names = FALSE)
cat("Table 4 saved. ANOVA F =", round(aov_dur_tidy$statistic[1], 2),
    "p =", format(aov_dur_tidy$p.value[1], scientific = TRUE), "\n")
```

[H] Figure 4: Track Duration Boxplot by Genre:

```r
selected_genres <- c(
  "grindcore", "children", "disney", "sleep",
  "classical", "jazz", "hip-hop", "edm", "dance", "country",
  "pop", "rock", "r-n-b", "indie", "soul", "blues",
  "techno", "chicago-house", "detroit-techno", "minimal-techno"
)

genre_meds <- df %>%
  filter(track_genre %in% selected_genres, duration_min < 10) %>%
  group_by(track_genre) %>%
  summarise(med = median(duration_min), .groups = "drop") %>%
  arrange(med)

df_dur_plot <- df %>%
  filter(duration_min < 10, track_genre %in% selected_genres) %>%
  mutate(track_genre = factor(track_genre, levels = genre_meds$track_genre))

genre_display_labels <- c(
  "grindcore"      = "Grindcore",    "children"      = "Children",
  "disney"         = "Disney",       "sleep"         = "Sleep",
  "classical"      = "Classical",    "jazz"          = "Jazz",
  "hip-hop"        = "Hip-Hop",      "edm"           = "EDM",
  "dance"          = "Dance",        "country"       = "Country",
  "pop"            = "Pop",          "rock"          = "Rock",
  "r-n-b"          = "R&B",          "indie"         = "Indie",
  "soul"           = "Soul",         "blues"         = "Blues",
  "techno"         = "Techno",       "chicago-house" = "Chicago House",
  "detroit-techno"= "Detroit Techno","minimal-techno"= "Minimal Techno"
)

levels(df_dur_plot$track_genre) <- genre_display_labels[levels(df_dur_plot$track_genre)]

p_dur_box <- ggplot(df_dur_plot, aes(x = track_genre, y = duration_min)) +
  geom_boxplot(fill = "#9ecae1", color = "#2171b5",
               outlier.size = 0.5, outlier.alpha = 0.3, linewidth = 0.5) +
  stat_summary(fun = mean, geom = "point", shape = 18,
               color = "firebrick", size = 2) +
  labs(title = "Distribution of Track Duration by Genre",
       subtitle = "Ordered by median duration; red diamond = mean; tracks > 10 min excluded",
       x = NULL, y = "Duration (minutes)") +
  theme_classic(base_size = 11) +
  theme(axis.text.x   = element_text(angle = 40, hjust = 1, size = 9.5),
        plot.title    = element_text(face = "bold", size = 12),
        plot.subtitle = element_text(size = 9, color = "gray40"))
```

[I] Figure 5: Mean Popularity by Speechiness Bin:

```r
df$speech_bin <- cut(df$speechiness, breaks = seq(0, 1, by = 0.05),
                     include.lowest = TRUE)

speech_means <- df %>%
  group_by(speech_bin) %>%
  summarise(mean_pop = mean(popularity), n = n(), .groups = "drop") %>%
  filter(!is.na(speech_bin))

speech_means$bin_mid <- seq(0.025, 0.975, by = 0.05)[1:nrow(speech_means)]

p_speech_bin <- ggplot(speech_means, aes(x = bin_mid, y = mean_pop)) +
  geom_col(aes(fill = mean_pop), width = 0.045, alpha = 0.9) +
  scale_fill_gradient(low = "#a1d99b", high = "#00441b",
                      name = "Mean\nPopularity") +
  geom_smooth(method = "loess", color = "firebrick", se = FALSE,
              linewidth = 1.2, span = 0.6) +
  labs(title = "Mean Track Popularity by Speechiness Bin",
       subtitle = "Bin width = 0.05; LOESS curve overlaid in red",
       x = "Speechiness", y = "Mean Popularity") +
  theme_classic(base_size = 12) +
  theme(plot.title    = element_text(face = "bold"),
        plot.subtitle = element_text(size = 9, color = "gray40"))
```

[J] Table 4: Overall Two-Sample t-Test:

```r
t_all <- t.test(popularity ~ explicit, data = df, var.equal = FALSE)

t_all_df <- data.frame(
  Group           = c("Non-Explicit (False)", "Explicit (True)"),
  n               = as.integer(table(df$explicit)),
  Mean_Popularity = round(tapply(df$popularity, df$explicit, mean), 3),
  SD              = round(tapply(df$popularity, df$explicit, sd), 3)
)

write.csv(t_all_df, "table5_overall_explicit_summary.csv", row.names = FALSE)

t_result_df <- data.frame(
  Statistic = c("t-statistic", "Degrees of Freedom", "p-value",
                "95% CI Lower", "95% CI Upper"),
  Value = c(round(t_all$statistic, 4),
            round(t_all$parameter, 2),
            format(t_all$p.value, scientific = TRUE),
            round(t_all$conf.int[1], 4),
            round(t_all$conf.int[2], 4))
)
```

[K] Table 5: Genre-Level Welch t-Tests:

```r
selected_genres_exp <- c("hip-hop", "pop", "classical", "country", "comedy", "gospel")

genre_ttest_list <- lapply(selected_genres_exp, function(g) {
  sub      <- df[df$track_genre == g, ]
  n_exp    <- sum(sub$explicit)
  n_nonexp <- sum(!sub$explicit)
  if (n_exp < 2 | n_nonexp < 2) return(NULL)

  tt    <- t.test(popularity ~ explicit, data = sub, var.equal = FALSE)
  means <- tapply(sub$popularity, sub$explicit, mean)

  data.frame(
    Genre            = g,
    n_NonExplicit    = n_nonexp,
    n_Explicit       = n_exp,
    Mean_NonExplicit = round(means["FALSE"], 3),
    Mean_Explicit    = round(means["TRUE"], 3),
    Difference       = round(means["TRUE"] - means["FALSE"], 3),
    t_statistic      = round(tt$statistic, 4),
    df               = round(tt$parameter, 2),
    p_value          = round(tt$p.value, 6),
    CI_lower         = round(tt$conf.int[1], 3),
    CI_upper         = round(tt$conf.int[2], 3),
    Significant      = tt$p.value < 0.05
  )
})
```